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Measurement of RBC deformation and velocity in capillaries in vivo

Jae Hong Jeong [a,*], Yasusiko Sugii [a], Motomu Minamiyama [b], Koji Okamoto [a]

[a] *Department of Quantum Engineering and Systems Science, the University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo, 113-8654, Japan*
[b] *Faculty of Health Science, Hiroshima International University, 555-36 Gakuendai, Kurose, Kamo-gun, Hiroshima 724-0695, Japan*

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Abstract

Red blood cells (RBC) become deformed while flowing through capillaries. We captured images of blood flow in capillaries and of RBC in the rat mesentery using a high-speed camera at 2000 frames/s and then directly measured and estimated the deformation and velocity of RBC in a non-uniform capillary. The distribution of the capillary diameter was determined by image processing. We applied a deformation index and simple modeling to observe RBC deformation in capillaries. The average capillary diameter was approximately 6.2 μm, and the average velocity of RBC was about 1.85 mm/s. The average deformation index of RBC in the capillary was about 1.55. The present results showed that RBC in capillaries generally assume a specific shape depending on external forces such as the velocity of the blood flow and capillary diameter in vivo.

Keywords: Red blood cell; Deformation; Capillary; Microcirculation

Introduction

Blood flow in the microcirculation plays an important role in maintaining healthy tissues and organs by delivering oxygen and nutrients. Several investigators have examined blood rheology in vivo and in vitro (Baskurt and Meiselman, 2003; Goldsmith, 1986; Sugii et al., 2002). Blood can be considered as a two-phase fluid consisting of blood cell components such as red blood cells (RBC), white blood cells (WBC) and platelets suspended in an aqueous solution of organic substances, proteins and salts called plasma. The motion of RBC in capillaries is an important factor in blood rheology since viscosity closely depends on the ability of RBC to deform. When RBC flows through capillaries, they assume parachute- or jellyfish-like shapes in response to deforming forces such as intracellular fluid and the RBC membrane (Skalak and Branemark, 1969; Gaehtgens et al., 1980; Bagge et al., 1980). The process of RBC deformation is thought to facilitate gas transfer by increasing the area of the RBC surface in contact with the capillary endothelium.

The elastic deformability of RBC has been measured by means such as rheoscopy, ektacytometry, and micropipettes (Chien, 1977) and directly observed in a transparent capillary model (Tracey et al., 1995; Sutton et al., 1997; Tsukada et al., 2001; Gaehtgens et al., 1980; Bagge et al., 1980). The deformation of RBC flowing at varying velocity into a capillary model consisting of microchannels of varying width has been measured using a high-speed camera system. However, since the properties of the wall surface and cross-sectional shape in vitro considerably differ from those of vessels and capillaries in vivo, RBC motion should be measured in real-time capillaries in a live animal.

Here, we used an intravital microscope and a high-speed camera system to quantify RBC deformation and velocity in rat mesenteric capillaries in vivo and analyzed relationships among the deformation index, capillary diameter, and RBC velocity.

Materials and methods

Experimental system

The deformation and velocity of RBC were examined in rat mesenteric capillaries in vivo. Female Wistar rats (2 months old) were anesthetized with alpha-chloralose (30 mg/kg) and urethane (0.75 g/kg) subcutaneously and with pentobarbital sodium (10 mg/kg), and a life support system was installed to retain the heartbeat and respiration. The mesentery was placed on the stage of a

* Corresponding author. Fax: +81 3 5841 8627.
E-mail address: jeong@utnl.jp (J.H. Jeong).

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heated (37°C) inverted-type microscope (Nikon Diaphot TMD). Blood flow images in mesenteric capillaries were recorded for 1.0 s at 2000 frames/s at a resolution of 1024 × 1024 pixels in 8-bit grayscale using a high-speed CMOS camera system (Photron Fastcam APX) mounted on the microscope and an objective lens (CFI Fluor 40×, Nikon, Japan) with a numerical aperture (NA = 0.85). Since the depth of field was about 0.7 μm, corresponding to 10% of the capillary diameter, the vertical spatial resolution was sufficient. The images were directly saved to the hard disk of a personal computer. The measurement region was illuminated with a halogen lamp.

Blood flow images

Fig. 1 shows an instantaneous image of RBC flowing in a mesenteric capillary of a live rat. The size of the recorded region was approximately 176 × 176 μm^2. One pixel was about 0.172 μm. The focal plane was set at the center of the capillary in the direction of depth. The capillary was bifurcated at about x = 87 μm and y = 25 μm as shown in Fig. 1. White arrows indicate the direction of blood flow. The capillary diameters in the left and right side of the image were about 10 and 6 μm, respectively. The density of RBC was higher in the left, than in the right capillary. Since the capillary in the right enclosed by a dotted line in the figure was smaller and RBCs were clearly distinguishable, we analyzed the deformation of 10–15 RBC per image and velocity at this site, which was about 150 μm long and about 6 μm in diameter.

Estimation of capillary diameter

The capillary diameter in vivo was not constant due to the shape of the endothelium (Vink and Duling, 1996). We therefore estimated the capillary diameter by image processing the blood images. The mesentery itself was stable over the entire image, allowing the position of the capillary to be regarded as constant. We also confirmed the temporal absence of vasomotion, meaning that the diameter of capillary was also constant. Although the real cross-section of vessel was elliptical shape, in order to estimate the blood flow rate, we assumed that the cross-section of the capillary was round. Since the interior diameter of capillary was difficult to be directly determined in vivo images, we assumed that the interior diameter of capillary (D_c) was evaluated by the width of the RBC path. Here, the RBC path indicated the passage integration of each RBC in the capillary. This assumption was applicable to the small capillary whose diameter was less than that of resting RBC (about 7–8 μm) because the error of estimation was less than the thickness of the endothelial surface layer (ESL; about 0.4–0.5 μm) of the capillary (D_c < 8 μm).

In order to obtain one image of the RBC path from all images, we eliminated background noise from the respective original images using a local averaging method because the local intensity of the images easily interfered with the sequential images in vivo. Fig. 2(a) shows that the image intensity of the RBC path was lower at the narrow diameter of the capillary because the velocity of RBC was faster at this point.

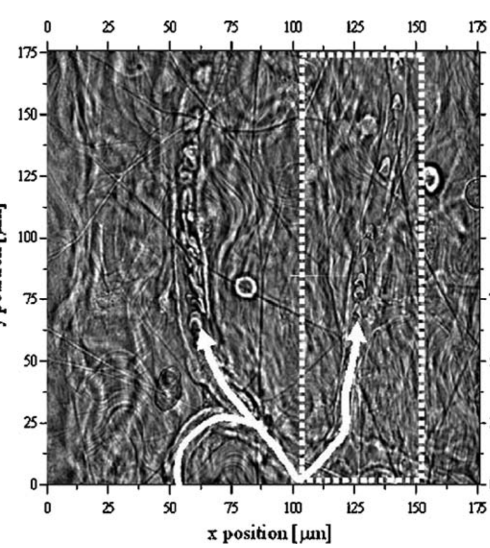

Fig. 1. Blood flow image in rat mesenteric capillary. Arrow shows direction of blood flow and dotted rectangle indicates capillary of interest.

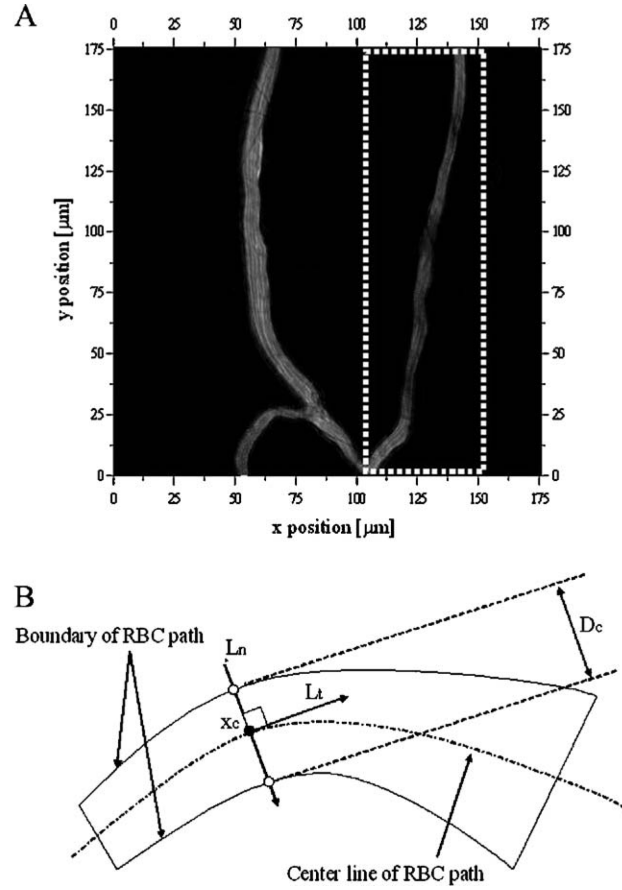

Fig. 2. (a) Image of RBC path in capillary. (b) Schematic diagram for estimation of interior diameter of capillary. L_t and L_n respectively indicate tangential and normal lines at x_c on center line of capillary.

We used a geometric approach to estimate the diameter of the capillary on the image of the RBC path. We established boundaries in the image by Laplacian edge detection. From the boundary data of the RBC path, a center line (L_c) of the path was calculated using a thinning algorithm. Fig. 2(b) shows a schematic estimation of the capillary interior diameter. The diameter of the RBC path at one point (x_c) on the center line (L_c) of the path was determined using two points where the normal line (L_n) met the boundary of the RBC path.

Estimation of RBC velocity and deformation index

In order to evaluate position and shape of RBC, RBC was extracted from the image using the local averaging method. Among 69 RBC passed along the capillary for 0.38 sec, 31 RBC were selected for analysis. The deformation index (DI) was used in the previous study (Tsukada et al., 2001) to evaluate the deformability of RBC in microchannel capillary. Definition of DI is as follows:

$$\mathrm{DeformationIndex(DI)} = \frac{L}{D} \quad (1)$$

where D is the diameter and L is the length of a deformed RBC in capillary.

In case of in vitro experiment (Tsukada et al., 2001) using a microchannel with straight path and constant rectangle cross-section, it is simple to decide the length and diameter of a deformed RBC because individual RBC almost deforms symmetrically along the flow direction. Since actually a capillary has non-uniform diameter and there are many curvature in the capillary wall, it is difficult to constantly keep the shape of RBC in capillary.

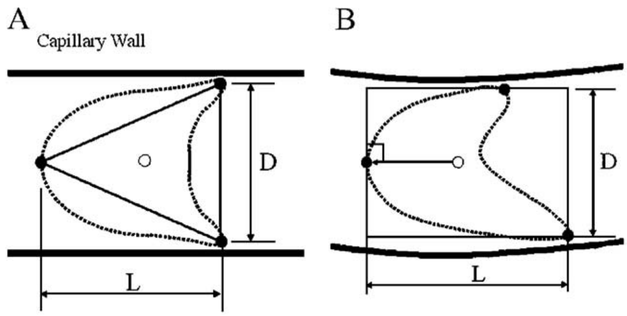

Fig. 3. Schematic diagrams of triangle configuration of RBC in capillary to simultaneously measure RBC deformation and velocity. Ideally – (a) and actually – (b) deformed RBC in capillary. *D* and *L*, diameter and length respectively, of deformed RBC in capillary.

We estimated RBC velocity and deformation index by fitting a triangle configuration of one head and two tail points to the RBC as shown in Figs. 3(a) and (b). Three points were selected by direct measurement from the images. Velocity was evaluated as displacement between the centroid of the three points on the RBC in the first and second images divided by time interval (0.5 ms) between the two successive images.

When the RBC was asymmetrically deformed in Fig. 3(a), these parameters could be easily determined using three points on each RBC instantaneously. Fig. 3(b) shows that RBC deform asymmetrically in vivo. In this case, we estimated RBC deformation in the capillary by drawing a rectangle that included the three points in Fig. 3(b) and positioned in the same direction as the blood flow.

Results and discussions

Capillary diameter

We defined the capillary axis (x_c) from the inlet $x_c = 0$ µm to the outlet $x_c = 160$ µm as shown in Fig. 4 (a). Fig. 4(b) shows the

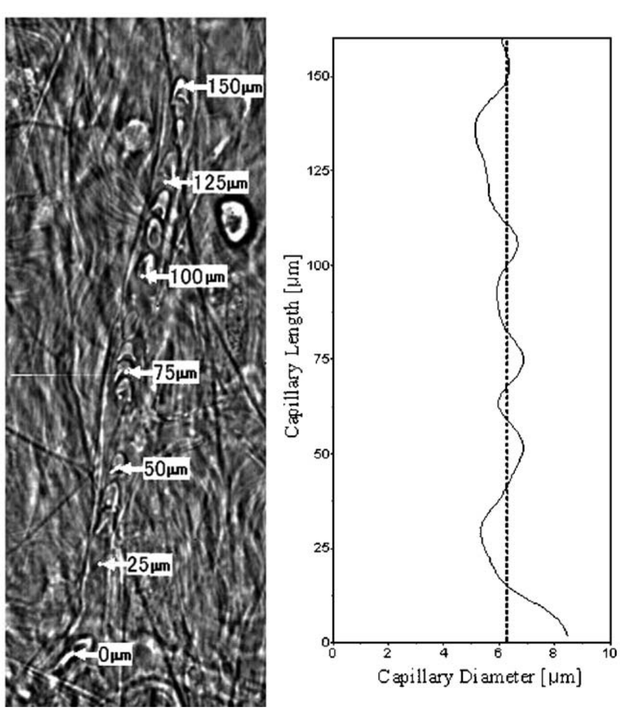

Fig. 4. (a) Capillary image. (b) Distribution of diameter along mesenteric capillary. Dotted line, average capillary diameter of 6.2 µm.

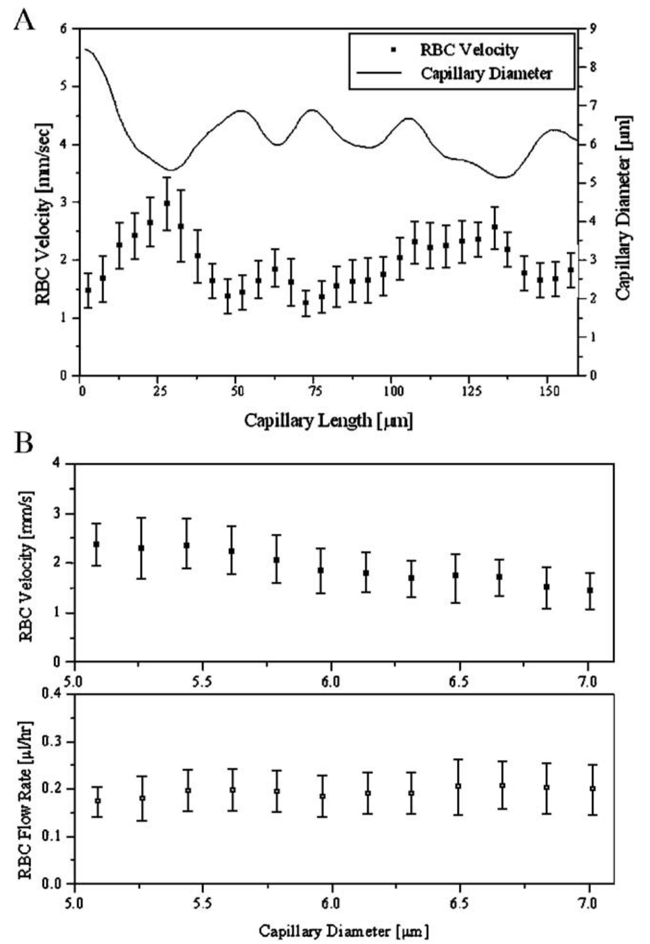

Fig. 5. (a) Distribution of RBC velocity in capillary. (b) Relationships of RBC velocity and flow rate, and capillary diameter. The error bar in the figure indicates the standard deviation.

width of the RBC path measured from the RBC path image (Fig. 2a). The average capillary diameter and standard deviation were 6.2 µm and 0.67 µm, respectively. The largest diameter was about 8.5 µm around the inlet ($x_c = 0$ µm), and the smallest was about 5 µm (about $x_c = 30$ µm and $x_c = 130$ µm). These results show that the capillary diameter changed in vivo along its length due to the shape of the endothelium. The width of the RBC path was smaller than the anatomic diameter estimated by the bright-field image. The difference between the obtained and the anatomic capillary diameter was about 0.2 to 0.9 µm. However, the standard deviation in the diameter difference was about 0.4 µm, which was similar to the thickness of the endothelial surface layer (ESL) (Gretz and Buling, 1995). The diameter of the capillary must be estimated as this factor directly affects RBC deformation in capillaries. This was particularly obvious in vivo when measuring RBC distortion in capillaries with a diameter that was smaller than that of resting RBC. However, the interior boundary of the capillaries was unclear.

RBC velocity along capillaries

The velocity of RBCs was more affected by the shape and diameter of the capillary. The entire velocity distribution shown

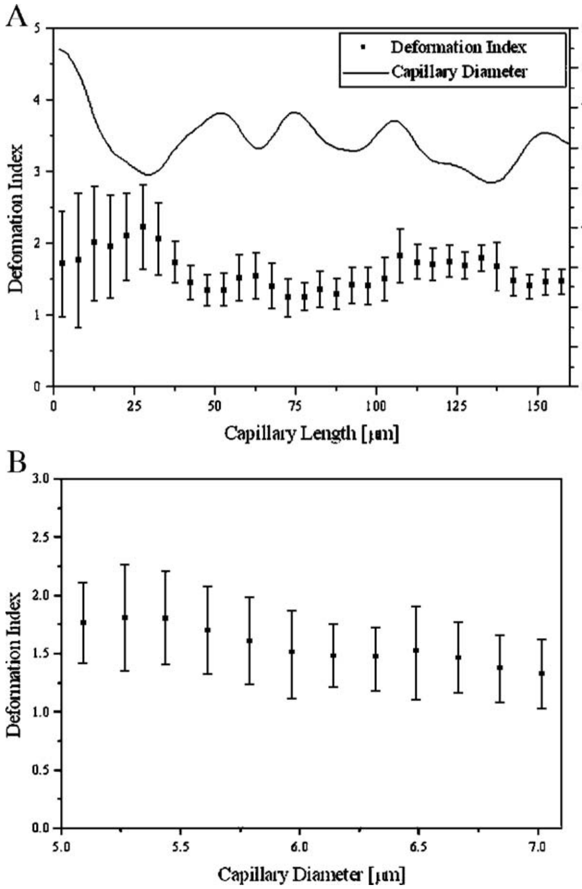

Fig. 6. (a) Distribution of RBC deformation index (DI) in capillary. (b) Relationship between deformation index (DI) and capillary diameter. The error bar in the figure indicates the standard deviation.

in Fig. 5(a) was inversely proportional to the capillary diameter. Average velocity and variance over the capillary were 1.85 and 0.4 mm/s, respectively. The maximal velocity of RBCs at x_c = 30 μm, which was the smallest diameter over the capillary, was about 2.93 mm/s, whereas the minimal velocity was approximately 1.25 mm/s at x_c = 73 μm. The velocity sharply increased from x_c = 25 μm to x_c = 50 μm by decreasing the capillary diameter and then decreased by increasing the diameter. The maximum standard deviation was 0.62 μm at x_c = 32.5 μm and minimum was 0.22 μm at x_c = 72.5 μm. The relationship between RBC velocity and capillary diameter is shown Fig. 5(b). The RBC velocity increased in the narrow part of the capillary. At a capillary diameter of 5 μm, RBC velocity was about 2.5 mm/s and at 7 μm about 1.5 mm/s. The velocity of RBC was faster in narrow than in wide capillaries. Fig. 5(b) shows the distribution of the RBC flow rate (Q_{RBC}) along the capillary estimated using the following equation:

$$Q_{\mathrm{RBC}} = \frac{1}{4}\pi D_c^2 V_{\mathrm{RBC}} \tag{2}$$

where D_c is the interior diameter of the capillary estimated by the width of RBC path (see 2.3) and V_{RBC} is the RBC velocity in the capillary, which has smaller diameter than the diameter of resting RBC. The average RBC flow rate over the range of capillary diameters appeared almost constant, $\bar{Q}_{\mathrm{RBC}} = 0.2\mu l/h$. This result indicated that even though RBC flow through non-uniform and narrow capillaries with a diameter that is smaller than that of resting RBC, the flow rate remains almost constant over the range of capillary diameters.

Deformation index of RBC

The deformation index (DI) was affected by RBC velocity as well as capillary diameter in the small capillary. Fig. 6(a) shows the distribution of the RBC deformation index. The distribution of the deformation index was particularly unstable in the inlet part of the capillary (x_c = 0–30 μm) where the diameter was larger than that of resting RBC. This means that the shape of the RBC can easily change and blood flow has not yet stabilized. The average deformation index was 1.55 in the capillary. After x_c = 50 μm, the RBC deformation index decreased with increasing the capillary diameter which was almost the same with deformed RBC diameter. When a RBC passes through a small diameter region of a capillary, it must elongate, thereby increasing the deformation index. The shape of RBC deformed asymmetrically in a small capillary because of the strength of shear stress between RBC membrane and the capillary wall.

Fig. 6(b) shows the relationship between the deformation index of tracked RBC and capillary diameter. The more the capillary diameter decreased, the more the deformation index increased. At a capillary diameter of about D_c = 5.0 μm, the deformation index was around 1.8, whereas at D_c = 7.0 μm, the value was about 1.3.

Thus, geometrical changes in the capillary comprised one determinant that affected the deformation of RBC when flowing in non-uniform capillaries, a fact that cannot be ignored in vivo. In contrast, the microchannels and glass capillaries used for studies in vitro have uniform rectangular and circular cross-sections, respectively (Sutton et al., 1997; Tsukada et al., 2001; Tracey et al., 1995; Bagge et al., 1980; Gaehtgens et al., 1980). However, our observations in vivo (Fig. 1) confirmed that cross-sections of a single capillary were variable.

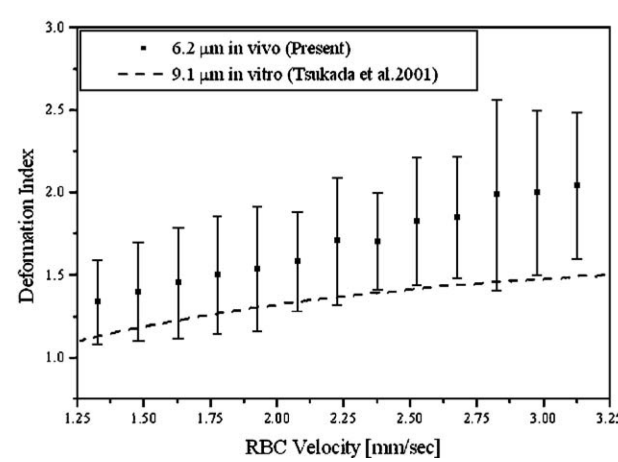

Fig. 7. Relationship between RBC velocity and deformation index. Dotted line, fitting data of deformation index of Tsukada et al. (2001) using microchannels in vitro. The error bar in the figure indicates the standard deviation.

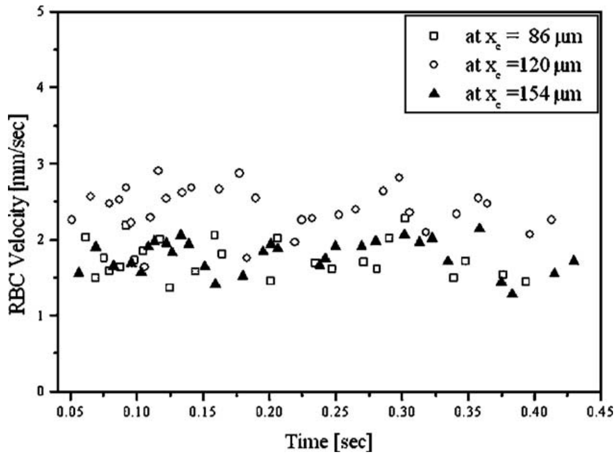

Fig. 8. Velocity distribution of RBC at three positions. x_c = 86, 120 and 154 μm; D_c = 6.0, 5.6, and 6.3 μm.

Relationship between deformation index and velocity

The deformation index of RBC in the capillary generally increases with increasing RBC velocity as shown Fig. 7. Fig. 7 shows the relationship between the RBC velocity and deformation index in the capillary. The dotted line shows the fitting data of Tsukada et al. (2001) describing the erythrocyte deformation index in vitro using microchannels 9.1 μm wide and 5.3 μm deep. They had used an exponential function fitting to predict the relationship between deformation index and velocity of deformed RBC. The result of this study shows that the relationship between RBC velocity and deformation index was qualitatively similar with Tsukada's result because the deformation index was affected by the velocity of plasma and capillary diameter both in vivo and in vitro experiment. However, the deformation index in the present study was higher than that of Tsukada's result because the diameter of the capillary in this study was smaller. In the small capillary, generally the diameter of deformed RBC (D) decreases and the length of deformed RBC (L) increases, thereby the deformation index (Eq. (1)) increases as a result.

The present study attempted to measure the interior diameter of, and to estimate the RBC velocity and deformation index during passage through, a capillary in vivo. In contrast to the many measurements of RBC deformability and deformation in vitro, few studies have examined alterations in these properties in a narrow capillary in vivo. Some studies on RBC in narrow glass tubes and microchannels showed that they appeared to assume axisymmetric parachute shapes (Hochmuth et al., 1970; Bagge et al., 1980; Tsukada et al., 2001) because of the standardized flow conditions prevailing in an environment with smooth inner walls, a constant inner diameter and no bends or bifurcations. The shape of RBC in our study appeared more variable and uneven compared with that in vitro. Furthermore, the RBC quickly became essentially axisymmetric as they entered a narrow capillary. This shows that the three dimensionality of RBC in narrow capillaries in vivo is far more variable than that in narrow glass tubes and microchannels.

Velocity fluctuation

Several studies using micro-PIV technique (Santiago et al., 1998) to measure the velocity distribution in blood vessels with a diameter of 20–30 μm showed the presence of a heartbeat in the vessels (Sugii et al., 2002; Nakano et al., 2003). In contrast to these results, the rat heartbeat (cardiac cycle of 6–7 Hz) in our study did not affect events in capillaries whose diameter was less than the diameter of resting RBC (about 8 μm). Fig. 8 shows the distribution of RBC velocity at three positions along the capillary: x_c = 86, 120 and 154 μm and D_c = 6.0, 5.6 and 6.3 μm. This result shows that the velocity distribution of RBC in a narrow capillary was affected by geometric properties such as the capillary diameter as well as cell–cell interactions between RBC more than by the heartbeat.

Conclusion

We examined the deformation and velocity of RBC in the capillary with a diameter that was smaller than that of resting RBC in vivo. We evaluated the interior diameter of the capillary using the measurement of the width of the RBC path. A high-speed camera system together with a triangle configuration being applied to the RBC allowed simultaneous measurements of the RBC deformation index and velocity. The shape of RBC almost deformed asymmetrically in a narrow capillary of the rat mesentery. This study also showed that flowing RBC entering the capillary assumed a parachute-like shape with changes in the capillary diameter. Deformation of the RBC was rapidly affected by a variability of capillary diameter and the RBC velocity. Furthermore, the velocity distribution of RBC in the capillary was affected by the capillary diameter and blood cell interaction more than by heartbeat.

References

Bagge, U., Branemark, P., Karlsson, R., Skalak, R., 1980. Three-dimensional observations of red blood cell deformation in capillaries. Blood Cells 6, 231–237.

Baskurt, O.K., Meiselman, H.J., 2003. Blood rheology and hemodynamics. Semin. Thromb. and Hemostasis 29 (5), 435–450.

Chien, S., 1977. Principles and techniques for assessing erythrocyte deformability. Blood Cells 3, 71–99.

Gaehtgens, P., Duhrssen, C., Albrecht, K.H., 1980. Motion, deformation and interaction of blood cells and plasma during flow through narrow capillary tubes. Blood Cells 6, 799–812.

Goldsmith, H.L., 1986. The microrheology of human blood. Microvas. Res. 31, 121–142.

Gretz, J.E., Buling, B.R., 1995. Measurement uncertainties associated with the use of bright field and fluorescence microscopy in the microcirculation. Microvasc. Res. 49, 134–140.

Hochmuth, R.M., Marple, R.N., Sutera, S.P., 1970. Capillary blood flow: I. Erythrocyte deformation in glass capillaries. Microvasc. Res. 2, 409–419.

Nakano, A., Sugii, Y., Minamiyama, M., Niimi, H., 2003. Measurement of red cell velocity in microvessels using particle image velocimetry (PIV). Clin. Hemorheol. Microcirc. 29, 445–455.

Santiago, J.G., Werely, S.T., Meinhart, C.D., Beebe, D.J., Adrian, R.J., 1998. A particle image velocimetry system for microfluidics. Exp. Fluids 25, 316–319.

Skalak, R., Branemark, P.I., 1969. Deformation of RBCs in capillaries. Science 164, 717–719.

Sugii, Y., Nishio, S., Okamoto, K., 2002. In vivo PIV measurement of RBC velocity field in microvessels considering mesentery motion. Physiol. Meas. 23, 403–416.

Sutton, N., Tracey, M.C., Johnston, I.D., Greenaway, R.S., Rampling, M.W., 1997. A novel instrument for studying the flow behaviour of erythrocytes through microchannels simulating human blood capillaries. Microvasc. Res. 53, 272–281.

Tracey, M.C., Greenaway, R.S., Das, A., Kaye, P.H., Barnes, A.J., 1995. A silicon micromachined device for use in blood cell deformability studies. IEEE Trans. Biomed. Eng. 42 (8), 751–761.

Tsukada, K., Sekizuka, E., Oshio, C., Minamitani, H., 2001. Direct measurement of erythrocyte deformability in diabetes mellitus with a transparent microchannel capillary model and high-speed video camera system. Microvasc. Res. 61, 231–239.

Vink, H., Duling, B.R., 1996. Identification of distinct luminal domains for macromolecules, erythrocytes, and leukocytes within mammalian capillaries. Circ. Res. 79, 581–589.